

Generation Interconnection System Impact Study Report

For Queue Project Q12-0107

Dresden Generating Station 100 kV

October 30, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0107. The Interconnection Customer proposes a 74.7 MW natural gas combined cycle facility interconnecting at the Dresden Generating Station 100 kV substation in the Illinois area. The request was received on February 20, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 80 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,060,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Dresden Sta – Dresden Generating Station 100.0 kV Circuit 2: 113.3% loading (149 MVA) following the outage of the Dresden Sta – Dresden Generating Station 100.0 kV #1; DFAX = 7.8%.

The Dresden Station – Dresden Generating Sta 100.0 kV #1 is overloaded to 110.5% of its 153 MVA emergency rating for the loss of the Dresden Sta – Dresden Generating Station 100.0 kV Cir. 2. The project contributes 7.8% DFAX (5.8 MW) to this overload.

Belridge 1A – Lake Mirror Substation 100.0 kV: 109.1% loading (153 MVA) following the outage of the Boca Raton 230/100kV Transformer Bank; DFAX = 6.4%.

For the loss of Bogard – Las Piedras Substation 100.0 kV Line 2, the Las Piedras – Bogard 100.0 kV loads to 107.8% of its 145 MVA rating. Project DFAX: 5.3%.

Kendall County Generation Facility – Wauneta 100.0 kV: 104.9% loading (143 MVA) following the outage of the Kendal County Generation Facility – Wauneta 100.0 kV; DFAX = 16.6%.

The Kendall County Generation Facility – Wauneta 100.0 kV Cir. 2 is overloaded to 104.9% of its 143 MVA emergency rating for the loss of the Kendall County Generation Facility – Wauneta 100.0 kV #1. The project contributes 16.6% DFAX (12.4 MW) to this overload.

The Bogard – Las Piedras 100.0 kV #2 is overloaded to 104.4% of its 150 MVA emergency rating for the loss of the Las Piedras – Bogard 100.0 kV #1. The project contributes 5.3% DFAX (4.0 MW) to this overload.

The Wauneta – Locust 100.0 kV is overloaded to 104.2% of its 105 MVA emergency rating for the loss of the Wauneta – Otay Mesa Substation 100.0 kV line. The project contributes 18.9% DFAX (14.1 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dresden Generating Station – Dresden Station 100.0 kV Line 2	Rebuild line with higher-capacity conductor	\$786,000	100% (\$786,000)
Dresden Generating Station – Dresden Station 100.0 kV Line 1	Rebuild line with higher-capacity conductor	\$782,000	100% (\$782,000)
Belridge 1A – Lake Mirror Substation 100.0 kV	Reconductor line	\$1,148,000	100% (\$1,148,000)
Las Piedras – Bogard 100.0 kV Line	Reconductor line	\$606,000	100% (\$606,000)
Kendall County Generation Facility – Wauneta 100.0 kV	Reconductor line	\$1,940,000	100% (\$1,940,000)
Kendall County Generation Facility – Wauneta 100.0 kV Cir. 2	Reconductor line	\$1,985,000	100% (\$1,985,000)
Bogard – Las Piedras 100.0 kV Circuit 2	Reconductor line	\$592,000	100% (\$592,000)
Wauneta Substation – Locut 100.0 kV	Reconductor line	\$626,000	35% (\$221,000)
	Total allocated to project		\$8,060,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.